Predicting Churn: Leveraging Machine Learning to Bolster Customer Retention

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Dataset:
For our project, we are selecting the customer churn dataset. This dataset consists of 3300 observations of 21 different variables, including one binary class attribute indicating whether or not a customer is predicted to churn in the near future.

Project Approach:
In order to provide the company with an accurate recommendation for how to address potential customer churn in the near future, we plan on employing the following methods and strategies:

- Read, clean and analyze the data using Python
- Perform Exploratory Data Analysis (EDA) by creating histograms for attributes of concern
- Utilize SAS to implement decision tree and Naïve Bayes classifications
- Employ a 10-fold cross validation data split strategy to accurately and comprehensively evaluate the model's performance across various subsets of the data
- Implement SQL integration using SQLite to store the cleaned dataset and run queries to analyze customer churn

Workload Distribution:

Group Member	List of Tasks
Jocelyn	- Identify and replace missing values in the dataset - Calculate the max, min, mean and std. deviation for each of the 21 attributes - Identify and account for outliers - Analyze the distributions of numerical attributes

	- Convert categorical variables into a suitable format for modeling - Plot histograms for attributes of concern and analyze whether they have any influence on the churn attribute
Henry	- Implement a decision tree classifier and evaluate results - Implement a Naïve Bayes Classifier and evaluate results - Employ a 10-fold cross-validation data split strategy for reliable model performance measurement - Compare decision tree and Naïve Bayes models based on accuracy, precision, recall, and F1-score - Generate tables and graphs to compare model results
Syed	- Document and summarize findings from data preparation and predictive modeling (classification) phases - Write conclusions and recommendations based off classification results - Implement SQL integration using SQLite (by storing the cleaned dataset and running queries to analyze customer churn) - Format and finalize the project report by integrating the conclusions and recommendations with tables, charts, and other forms of effective visualization